

BanglaWeatherBench: Monsoon-Stratified Evaluation of Zero-Shot Time-Series Foundation Models on Station Observations

Ismail Hossain Polas
Jahangirnagar University
ismailhossainpolash@gmail.com

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Abstract

Time-series foundation models promise that you can pretrain once and then forecast any series without fitting. The evidence comes almost entirely from mid-latitude, data-rich places, and often from reanalysis rather than from instruments. BanglaWeatherBench asks what that promise is worth in a tropical monsoon climate, measured against what stations actually recorded. It scores 8 zero-shot foundation models against 10 trained and statistical baselines on 16 daily tasks over 170,571 rolling-origin windows, on 35 Bangladesh Meteorological Department stations. Three controls make the result readable: 32 European stations from the same network; NASA POWER reanalysis at the same 77 coordinates; and a context-gap control applied to models and baselines alike. We resample at the forecast origin, because overlapping windows make ordinary tests overconfident. The result is mixed, and the nulls are the useful part. Foundation models are the strongest family, yet the best one is statistically indistinguishable from the best baseline in 29 of 48 comparisons. On station observations it wins 3 of 24; on reanalysis at those coordinates, 11 of the same 24. Against a day-of-year climatology on tropical rainfall nothing wins at all: not beyond a one-day lead at the stations, nor at any horizon on ten-day satellite rainfall. Reanalysis looks more predictable than the stations it stands in for in 78 of 108 comparisons, so benchmarks built on it flatter models. At a 30-day lead 4 of 8 models lose significantly more temperature skill in Bangladesh than a gap-matched control; roughly 60% of that gap is sparser station context, not climate. Skill also reverses inside the monsoon: strong in break spells, worse than climatology in active ones, when forecasts matter most. The whole leaderboard reproduces in 4.16 GPU-hours on a Tesla T4. Compute is not the barrier.

1 Introduction

A time-series foundation model is pretrained once on a large heterogeneous corpus, then asked to forecast an unseen series with no fitting, no feature engineering and no local expertise. If that promise holds where forecasting capacity is scarcest, it matters. Those are the places with the fewest stations, the fewest trained forecasters and the least compute, and they are often where a week of warning carries the largest consequences. A model that runs zero-shot on a single GPU is precisely the technology they lack.

Whether it holds there is an empirical question current benchmarks do not answer, because the evidence base is skewed in two ways at once. Geographically, general time-series suites treat weather as one domain among many and draw it overwhelmingly from mid-latitude sources [2, 13, 18, 29]; across seven widely used benchmark papers, the words “monsoon” and “Bangladesh” do not appear. Representationally, the weather benchmarks that *are* globally comprehensive score against reanalysis [27, 30]. Reanalysis is a physically consistent reconstruction, smoother than the station record it interpolates; it is smoothest exactly where stations are sparsest.

Observation-based evaluation exists for numerical and AI weather-prediction models [14, 16], but the corresponding question for zero-shot time-series models has not been asked.

Bangladesh makes the question concrete. Its weather is governed by the South Asian summer monsoon, whose defining feature, the alternation of active and break spells inside the rainy season, has no temperate analogue. The 35-station record we build on shows the range. Median July rainfall reaches 457 mm per station-month, against 0 mm in January. 67.6 % of station-days are dry, yet the wet-day 99th percentile is 143 mm. The 2016–2023 test period alone contains 432 cyclone station-days across 14 storms. The record that would support decisions here is thin. Over the same period the ten Bangladeshi stations in GHCN-Daily are about 65 % complete, with 2021 almost entirely absent. The European stations we use as a reference are 99 % complete. That asymmetry is not only a property of some pretraining corpus: it sits inside the model’s input window at forecast time.

We introduce **BanglaWeatherBench**, a benchmark designed so that its headline number is not the only thing it produces. It defines 16 daily forecasting tasks over 170,571 rolling-origin windows across four tracks. The forecast target is 35 Bangladesh Meteorological Department stations; beside them sit 10 co-located GHCN-Daily stations, 32 European GHCN-Daily stations as a temperate reference, and NASA POWER reanalysis at all 77 station coordinates. A seventeenth task covers CHIRPS dekadal rainfall over 64 districts, from an independent source. Three controls answer the confounds that would otherwise make a tropical result unreadable. The same-network temperate reference isolates climatic regime; the reanalysis track at identical coordinates isolates the effect of the scoring target. The gap-matched context control isolates data availability from climate, and we apply it to the trained baselines as well as to the foundation models. Scoring is stratified by monsoon phase and by active/break spell.

Two commitments shape what we are willing to claim. First, significance. A 7-day stride with a 30-day horizon produces heavily overlapping windows, so we resample at the forecast origin. We report Diebold–Mariano tests with HAC variances, moving-block bootstrap intervals, Friedman–Nemenyi ranks, and Holm correction across 48 comparisons. Many margins that look decisive as point estimates do not survive. Second, leakage: we audit each model’s pretraining corpus against its technical report. Leakage-free zero-shot evaluation is not itself novel [28]; here it is a precondition for reading a geographic contrast as a geographic effect.

Evaluating 8 zero-shot foundation models against 10 trained and statistical baselines yields a mixed result that we think is more useful than either a triumphal or a dismissive one.

1. **Foundation models lead, but most of the lead is not separable.** The best model beats the best baseline in 14 of 48 Holm-corrected comparisons, loses 5 and is indistinguishable in 29; on station observations it wins 3 of 24, all at a one-day lead, against 11 of 24 on reanalysis. The six leading models and PatchTST sit within one critical difference.
2. **Nothing beats climatology on tropical rainfall** beyond a one-day lead at the BMD stations, or at any horizon on dekadal rainfall, a concrete boundary on the “forecast anything” claim.
3. **Reanalysis flatters models** relative to the stations it represents, in 78 of 108 comparisons.
4. **The tropical temperature deficit is narrow and mostly about data.** At a 30-day lead 4 of 8 models lose significantly more skill than a like-for-like control; about 60 % of the raw penalty is explained by sparser context, and doubling the context does not remove the remainder.
5. **Skill reverses by regime:** strong in break spells, significantly worse than climatology in active spells, in 61 of 72 tests.

We state finding 4 carefully because it is easy to misreport. The evidence supports a specific claim: for temperature at long horizons, half the roster loses significantly more skill on Bangladeshi stations than trained baselines do, most of it traceable to sparser context. It does not support a blanket claim that foundation models fail in the Global South. Stated correctly it is also the more actionable result: much of the deficit is a data-availability problem that observation programmes and gap-tolerant architectures can address.

Contributions. (i) A released, leakage-tested benchmark for a tropical-monsoon climate, with splits, strata, a datasheet and per-source licensing. (ii) Controls rather than a leaderboard alone: a same-network temperate reference, a reanalysis track at identical coordinates, and a gap-matched control applied to models and baselines alike. (iii) A significance protocol matched to overlapping rolling-origin panels, applied uniformly. (iv) To our knowledge the first evaluation of current zero-shot time-series foundation models on South Asian station observations including precipitation, stratified by monsoon phase, with a pretraining-overlap audit. (v) Reproducibility at the compute scale of the places the benchmark is about: 4.16 GPU-hours on a Tesla T4.

2 Related work

General time-series benchmarks. Monash [13], GIFT-Eval [2], TFB [29] and TSFM-Bench [18] evaluate across many domains and report aggregate ranks. Weather appears as one domain among dozens, drawn mostly from mid-latitude sources, and none stratifies by climatic regime or geography. TIME [28] adds a strict leakage-free protocol over 50 fresh datasets and 12 foundation models, and likewise applies no geographic or climatic strata. Recent work has begun to document that aggregate metrics conceal regime-dependent failure [32] and that benchmark construction itself is a source of leakage [25]; our design follows both warnings.

Weather benchmarks. WeatherBench 2 [30] and ChaosBench [27] score against reanalysis; ExtremeWeatherBench [23] scores against event catalogues; WeatherReal [16] and WEATHER-5K [15] score against station observations. WEATHER-5K is the closest in spirit, with thousands of global stations at hourly resolution. But it carries neither precipitation nor humidity, the variables at issue here, and reports global aggregates rather than climatic strata. None of these evaluates time-series foundation models on a tropical station network.

Geographic disparity in AI forecasting. SAFE [22] stratifies AI weather-prediction models by country, subregion and income group, finding disparities across every attribute, but works on ERA5 at 1.5° resolution for upper-air fields only. Mozaffari et al. [26] argue qualitatively that AI risks amplifying a North–South divide without supplying measurements. Rowell Jr. and Kupssinskü [31] stratify GraphCast by climatic sub-region over Brazil against operational IFS analysis for T_{850} , Q_{850} and Z_{500} , excluding precipitation. Ours is the corresponding measurement for time-series foundation models, on station observations, including rainfall.

South Asian monsoon and AI models. Gupta et al. [14] assess seven global AI weather-prediction models against South Asian observations during the monsoon, reporting errors 15–45% larger than against reanalysis. That is the precedent for our observation-versus-reanalysis contrast, transferred here to a different model class. Luitel et al. [21] evaluate GraphCast over the Indian summer monsoon against ERA5 and IMERG, and Aitken et al. [1] build decision-oriented probabilistic onset forecasts for Indian farmers from AI weather models. All three concern gridded NWP-style emulators and onset or seasonal-mean skill rather than zero-shot station forecasting.

Precipitation-specific evaluation. A recent benchmark evaluates 17 supervised architectures for 0–6 hour GNSS-based precipitation nowcasting [34]. The task, horizon and model class all differ from ours, but it is the closest existing precipitation-focused time-series benchmark.

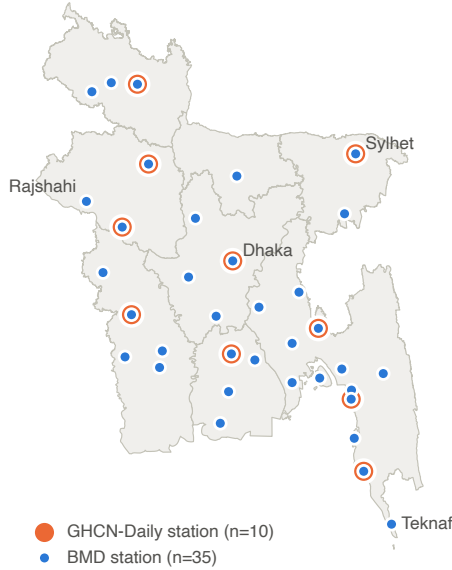
Bangladesh forecasting. The applied literature is substantial but uses per-station supervised models and releases no shared tasks: Joy et al. [17] is representative, training attention-enhanced LSTMs for temperature and rainfall. No published work evaluates pretrained foundation models on Bangladeshi station data, and no Bangladesh weather benchmark with fixed splits and a leaderboard exists.

Position. The intersection is, to our knowledge, unoccupied: current time-series foundation models, on tropical station observations including precipitation, stratified by monsoon regime, with a matched temperate control and a significance protocol. We claim novelty only at that intersection, and cite the neighbours above for each component individually.

3 BanglaWeatherBench

The benchmark is built around one observational target and three controls, each answering a confound that would otherwise make a tropical result unreadable.

a Bangladesh observation stations



b GHCN-Daily tracks: same WMO network, two climates

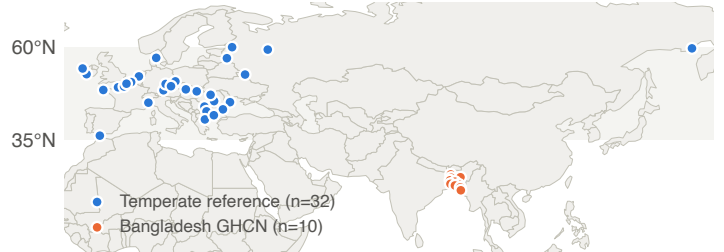


Figure 1: The three station networks. Left: 35 BMD stations across Bangladesh (the forecast target), with the 10 co-located GHCN-Daily Bangladesh stations marked. Right: the 32 mid-latitude European GHCN-Daily stations that form the temperate reference. Both station sets are drawn from instrument networks reporting to the same WMO standards, which is what allows climatic regime to be varied while the observing protocol is held fixed. Administrative boundaries for Bangladesh are the OCHA/HDX Common Operational Datasets (CC BY-IGO); world boundaries are Natural Earth (public domain).

3.1 Data tracks

BMD stations (target). Daily observations from 35 Bangladesh Meteorological Department stations, 1961–2023, covering rainfall (mm), temperature (°C), relative humidity (%) and bright sunshine (hours), released under CC-BY-4.0 [35]. The published record was already imputed by its authors using forward/backward filling, interpolation and manual edits, so we detect and flag suspect values rather than trusting the file: constant runs, perfectly linear runs, and values lying off the recording grid. Over the test period, flagged shares are 0.08 % (rainfall), 1.26 % (temperature), 1.75 % (sunshine) and 4.44 % (humidity); the worst single station-variable is 14.1 %, and only 11 of 140 station-variables exceed 5 %. A window is scored only if its target is complete *and* unflagged, so flagged values never enter a score.

GHCN-Daily Bangladesh (cross-check). Ten stations from the WMO-derived network [24], co-located with BMD sites, giving an independently curated view of the same weather. Over 2016–2023 these are about 65 % complete, with 2021 almost entirely absent. GHCN labels a rainfall total by the day the 24-hour accumulation begins and BMD by the day it ends; we correct the one-day offset only for cross-source comparison, never for per-track forecasting.

GHCN-Daily temperate (reference). Thirty-two mid-latitude European stations from the same network, selected for record length and completeness (about 99 % over the test period). Holding the instrument network, variables, protocol and scoring fixed isolates climatic regime from everything else. All 32 are European; none is in the United States, which matters for the leakage audit (Section 4).

NASA POWER (reanalysis). Reanalysis sampled at all 77 station coordinates, so that observation and reanalysis can be compared at identical places and dates rather than across different domains.

CHIRPS dekadal (independent source). Ten-day rainfall totals for 64 districts, 1981–2026, from a satellite-gauge product. Two administrative units are split across polygons sharing one identifier; we merge them with pixel-count weights rather than dropping either.

3.2 Tasks

Sixteen daily tasks are defined as the cross of tracks and variables: four BMD variables, two GHCN-Bangladesh (rainfall, mean temperature), two GHCN-temperate, and the four NASA POWER analogues at each of the three station sets. A seventeenth task covers CHIRPS dekadal rainfall at a six-dekad horizon. Tasks are evaluated separately; we never pool across tracks, because scaled errors are not comparable between a smooth reanalysis grid and a sparse station record.

3.3 Evaluation strata

Scoring is stratified by physically defined, externally validated regimes rather than by empirical clusters. Monsoon phases (pre-monsoon, onset, peak, withdrawal, dry) follow published Bangladeshi climatology, and active/break spells follow Ferdoushi et al. [9]: runs of at least three days in July–August where daily rainfall on the all-station mean series sits above or below half a standard deviation of the daily climatological cycle. Our labels reproduce that paper’s counts to within a few percent. Strata are derived from the training period only and are used for scoring alone; they are never model inputs.

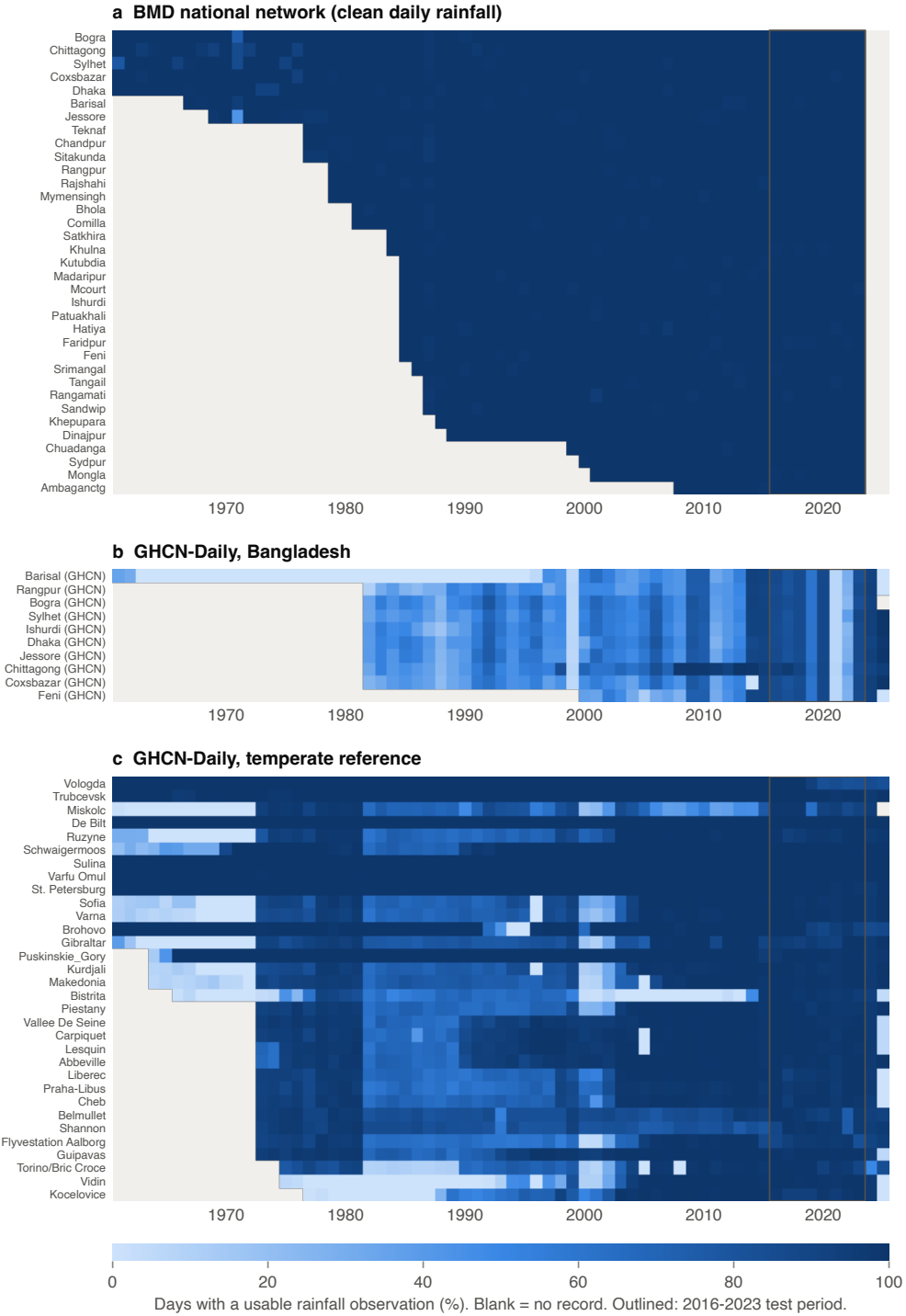


Figure 2: Usable observations per station-year, for the three station tracks. The 2016–2023 test period is outlined. The BMD target record is dense; the GHCN-Daily Bangladesh track is roughly 65 % complete over the test period with 2021 almost entirely absent, against 99 % for the temperate reference. This gap is a property of the observing system rather than of the climate, and it is what the gap-matched control in Section 4 exists to neutralise.

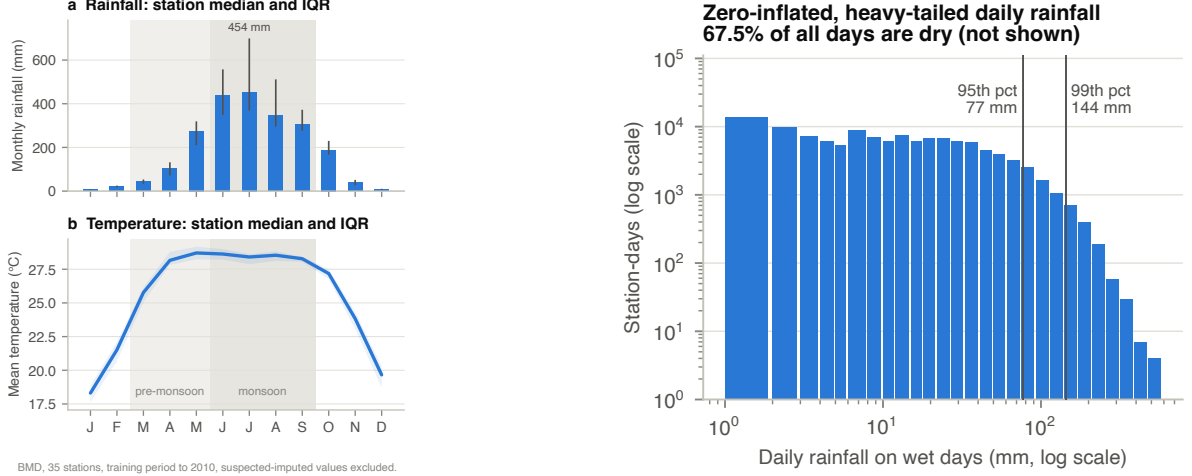


Figure 3: Two properties of the target record that shape the results. Left: the annual cycle of rainfall and temperature across the 35 BMD stations over the training period, showing the monsoon window the phase strata partition. Right: daily rainfall on wet days, log-log. Two thirds of station-days are dry and the wet tail runs past 500 mm, so a rainfall forecast is mostly a forecast of occurrence; this is the distribution against which climatology proves hard to beat (Section 5).

4 Evaluation protocol

4.1 Splits and windows

Temporal splits are fixed for every track: training up to 2010-12-31, validation 2011–2015, and a 2016–2023 test period common to all tracks. Anything estimated from data (climatologies, percentile thresholds, the spell reference cycle, and model fitting) uses the training split only.

Evaluation is rolling-origin with a 7-day stride and horizons of 1, 3, 7, 14 and 30 days. A window is scored when its 30-day target lies entirely inside the test split, is complete and unflagged, and at least 365 days of history precede the origin. Context is deliberately *not* required to be complete: a 90%-complete-context rule leaves zero admissible GHCN-Bangladesh rainfall windows, which would silently delete the sparsest and most representative part of the problem. Models therefore receive contexts containing real gaps, and each window records its context completeness so that skill can be stratified by it. This yields 170,571 daily windows and 18,112 dekadal windows.

4.2 Metrics

The headline metric is the continuous ranked probability score, a strictly proper scoring rule [12], computed here from nine predictive quantiles as twice the mean pinball loss. Point errors use each model’s predictive *median* and squared errors its mean. Mixing the two is not neutral on zero-inflated rainfall. Scoring climatology on its mean rather than its median changed its MASE at a 30-day lead from 0.58 to 0.82. It also made a quantile-loss network appear 25 % better than it is. Interval calibration is reported as coverage of the central 80 % interval.

Because scaled errors differ systematically between tracks, all cross-track comparisons use skill relative to the *same track’s* day-of-year climatology, estimated on the training split. A score of zero is climatology.

4.3 Significance

Neighbouring origins overlap heavily, so treating windows as independent samples would overstate confidence substantially; none of the reference forecasting benchmarks reports significance tests at all. We therefore make the forecast origin the unit of resampling, averaging stations within an origin so that cross-sectional correlation is absorbed rather than counted as extra samples.

Pairwise comparisons use Diebold–Mariano on the paired loss differential, with a Newey–West HAC variance at lag $h - 1$ and the Harvey–Leybourne–Newbold small-sample correction, referred to a t distribution. Interval estimates come from a moving-block bootstrap over origins, with blocks of 8 origins and 2,000 resamples. Paired series (a model, its climatology reference and the control baseline) are resampled with the same blocks. A difference-in-differences interval therefore carries the control’s uncertainty rather than treating it as fixed. Rankings use Friedman with the Nemenyi critical difference from the studentized range. All 48 task-and-lead comparisons are Holm–Bonferroni corrected, and we report the corrected values.

4.4 Controls

Gap-matched contexts. Bangladeshi contexts are 82–86% observed over the year preceding an origin, against 99–100% for the temperate reference (Figure 2). To separate climate from data availability we draw real missingness patterns from GHCN-Bangladesh windows and impose them, right-aligned and deterministically, on temperate contexts. Masking moves the temperate observed share to 0.808 for rainfall (Bangladesh reference 0.819) and 0.863 for mean temperature (reference 0.863). Crucially the mask is applied to the trained baselines as well as to the foundation models, so the control is like-for-like; masking only the model understates the remaining effect.

Leakage audit. Every model’s pretraining corpus was checked against its technical report rather than its model card. No corpus names GHCN-Daily, BMD or any Bangladesh source. The station corpora that do appear, USHCN and SubseasonalClimateUSA, are United States only, and our temperate reference is entirely European, so neither side of the tropical-versus-temperate contrast is contaminated. Leakage-free zero-shot evaluation is not itself novel [28]; what matters here is that it is a precondition for reading a geographic contrast as a geographic effect.

Context length. Because a sparse context is also a shorter effective context, the tropical contrast is repeated with 2,048 days of history in place of 1,024. The control is masked identically.

5 Results

We evaluate 8 zero-shot foundation models against 10 trained and statistical baselines on identical windows: Chronos-2 [4] and Chronos-Bolt [3], TimesFM 2.5 [7], Toto-2.0 [6], TiRex [5], Moirai-2 [19, 33], Granite TTM r2 [8] and Sundial [20]. All models see a 1,024-day context except TTM, whose daily variants cap at 512.

5.1 Foundation models lead, but most margins are not separable

Averaged over the 16 daily tasks, six of the eight foundation models outrank every trained baseline. Chronos-2 has the best mean rank at leads 7 and 30, 2.77 of 18 models overall; TiRex is best at lead 1. The strongest trained baseline is PatchTST, at 6.96 (Table 1). Friedman rejects equality decisively at every lead ($\chi^2 = 220.1, 215.5, 209.8$; $p < 10^{-34}$).

That ranking, however, does not license an ordering. The Nemenyi critical difference is 6.58. At leads 7 and 30, the six leading foundation models and PatchTST all fall within one critical difference of the best (Figure 4). The group separates clearly from the weaker baselines; positions within it do not.

Pairwise, the picture is more sober still. We compared the best foundation model with the best baseline for each task and lead, correcting across all 48 comparisons. The foundation model is significantly better in 14, worse in 5, and statistically indistinguishable in 29. Those wins concentrate at a one-day lead and on the smooth reanalysis tracks. Partitioned by track type, foundation models take 11 of 24 comparisons on reanalysis and only 3 of 24 on the stations themselves, all three of those at lead 1. Of 432 model-task-lead skill intervals, 272 exclude climatology.

Table 1: Main leaderboard over the 16 daily tasks. Ranks are Friedman average ranks across the tasks at each lead (1 is best, 18 models); *Overall* is the mean of the three. CRPSS is mean skill against the same track’s day-of-year climatology at a 7-day lead, so a positive value beats climatology and zero is climatology, averaged over the 8 station-observation tasks and the 8 reanalysis tasks separately. The Nemenyi critical difference is 6.58, so rank gaps smaller than that are not separable. Parameter counts apply to the zero-shot models; the baselines are fitted per series. TTM r2 is point-only and is scored on point skill.

	Params	Average rank (1 = best)				Mean CRPSS, lead 7	
Model	(M)	Lead 1	Lead 7	Lead 30	Overall	Station	Reanalysis
<i>Zero-shot foundation models</i>							
Chronos-2	119.5	4.19	2.12	2.00	2.77	+0.035	+0.128
TiRex	35.3	1.62	3.44	3.62	2.90	+0.025	+0.121
TimesFM 2.5	231.3	4.38	4.44	3.75	4.19	+0.024	+0.121
Toto-2.0	312.7	3.75	5.81	6.56	5.38	−0.015	+0.117
Moirai-2	11.4	4.69	6.31	7.06	6.02	+0.003	+0.110
Chronos-Bolt	205.3	5.81	5.75	6.62	6.06	+0.016	+0.108
Sundial	128.3	11.06	10.69	10.31	10.69	−0.088	+0.031
TTM r2	0.8	16.38	15.94	15.25	15.85	−0.374	−0.246
<i>Trained and statistical baselines</i>							
PatchTST	n/a	7.75	6.31	6.81	6.96	+0.009	+0.080
AutoARIMA	n/a	7.62	8.62	8.88	8.38	−0.062	+0.025
NHITS	n/a	8.12	8.94	8.88	8.65	−0.002	+0.023
LightGBM	n/a	12.88	8.62	8.25	9.92	−0.063	+0.042
DLinear	n/a	13.25	12.00	11.50	12.25	−0.126	−0.005
AutoETS	n/a	11.62	14.44	15.81	13.96	−0.255	−0.180
AutoTheta	n/a	12.94	15.25	15.81	14.67	−0.268	−0.188
<i>Naive references</i>							
Climatology	n/a	14.62	8.50	7.25	10.12	+0.000	+0.000
Persistence	n/a	12.38	16.69	17.38	15.48	−0.444	−0.278
Seasonal naive	n/a	17.94	17.12	15.25	16.77	−0.406	−0.348

5.2 Nothing beats climatology on tropical rainfall

All five baseline wins are a day-of-year climatology on rainfall at 7- and 30-day leads. On the 35-station BMD record no foundation model exceeds climatology beyond a one-day lead; the best any manages is -0.004 CRPSS at lead 30. On the independent CHIRPS dekadal task none beats it at any horizon. The closest is Chronos-2 given 28 years of context, which matches climatology to within 0.001 CRPSS (Table 2).

The claim needs one scope marker. On the smaller 10-station GHCN-Bangladesh track the best foundation model does hold positive point estimates at every lead: Toto-2.0 $+0.095$ at day

Critical-difference ranking: foundation models (blue) vs baselines (grey)

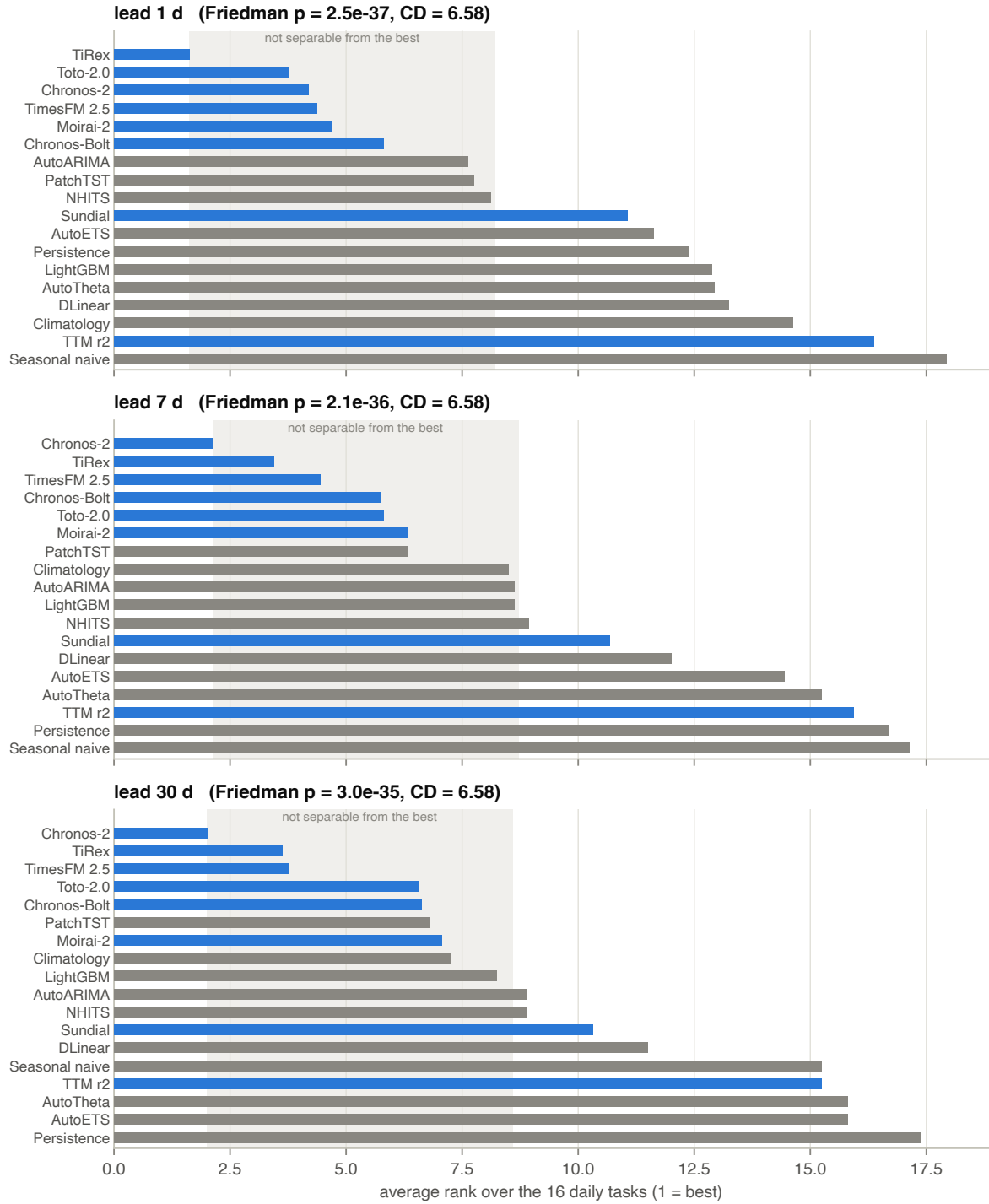


Figure 4: Average rank over the 16 daily tasks, with the shaded band marking one Nemenyi critical difference from the best model. Foundation models in blue, baselines in grey. The leading group is not internally separable.

1, Chronos-2 +0.019 at day 30. None of those margins survives correction ($p = 0.88, 1.00, 1.00$), so they are not separable from climatology either.

Table 2: CHIRPS dekadal rainfall, skill against a dekad-of-year climatology (higher is better; 0 is climatology). Headline context is 144 dekads; the final column gives the best result with all available history.

Model	Lead 1	Lead 2	Lead 3	Lead 6	Best, 1,024 dekads
Climatology	0.000	0.000	0.000	0.000	n/a
Chronos-2	-0.063	-0.083	-0.080	-0.086	-0.001
Toto-2.0	-0.086	-0.115	-0.114	-0.113	-0.004
TimesFM 2.5	-0.162	-0.150	-0.137	-0.135	-0.055
TiRex	-0.169	-0.219	-0.257	-0.323	-0.030
Moirai-2	-0.196	-0.290	-0.347	-0.504	-0.073
Persistence	-0.478	-0.635	-0.763	-1.158	n/a
TTM r2 (point only)	-0.682	-0.868	-1.002	-1.529	-0.846

5.3 Reanalysis flatters models against the stations it represents

Scoring the same models on NASA POWER and at the co-located stations, skill is significantly higher on reanalysis in 78 of 108 model-variable-lead comparisons (Figure 5). The gap is largest for humidity, significant at every lead for all nine models tested, and for rainfall, significant for all nine at leads 1 and 30. It is smaller for temperature. It is weakest for sunshine against radiation, two quantities that are not physically identical, which is why we do not lean on that row. The effect holds across model families, so it is a property of the target rather than of any architecture. A benchmark built on reanalysis therefore overstates two things at once: absolute skill, and the margin foundation models hold over baselines. At a 7-day lead the best foundation model’s advantage over the best trained baseline is +0.026 on observations, against +0.048 on reanalysis. This is the time-series counterpart of the critique Gupta et al. [14] make for AI weather-prediction models.

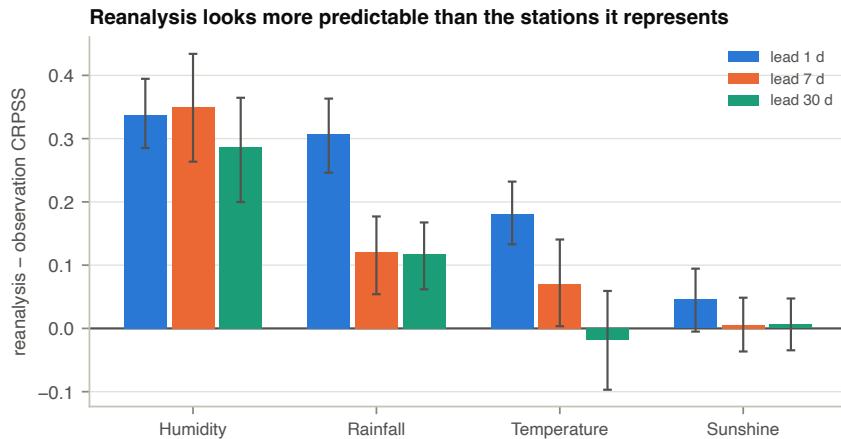


Figure 5: Skill on NASA POWER minus skill at co-located stations, with 95% moving-block intervals. Positive values mean the reanalysis looks more predictable than the observations it represents.

5.4 A long-lead tropical temperature deficit, mostly explained by sparse context

Against a trained-baseline control subjected to the same Bangladesh-like missingness, four of the eight foundation models retain a significant temperature penalty at a 30-day lead: TTM r2 (+0.187), Chronos-Bolt (+0.138), Chronos-2 (+0.060) and TiRex (+0.052). Once both sides face the same gaps, the median penalty across the roster falls from +0.112 to +0.043. Roughly 60 % of the raw effect is therefore attributable to sparser station context rather than to the tropics (Figure 6). At a one-day lead no model shows a significant penalty.

Two scope markers matter. First, masking only the model, as an ablation applied to the foundation models alone would, *understates* the effect, because the control also degrades on gappy contexts; the like-for-like comparison raises the median from +0.022 to +0.043 and returns TiRex to significance. Second, rainfall shows no systematic penalty at any lead. Where results are significant at lead 1 the sign is reversed, with four models performing relatively *better* in Bangladesh than at temperate stations. The single exception is Toto-2.0 at lead 7 (+0.028, CI 0.005 to 0.053).

Longer context does not dissolve what remains. Repeating the contrast with 2,048 days of history and the control masked identically leaves the median penalty at +0.052, unchanged from 1,024 days; Chronos-2 (+0.059) and TiRex (+0.052) stay significant, while TimesFM’s penalty, which was genuinely context-related, disappears. An unmasked comparison would have suggested the opposite, because without masking the median halves from +0.080 to +0.043 as context doubles. That apparent context effect is mostly the missingness effect in disguise.

Tropical temperature penalty, and how much of it is missing context

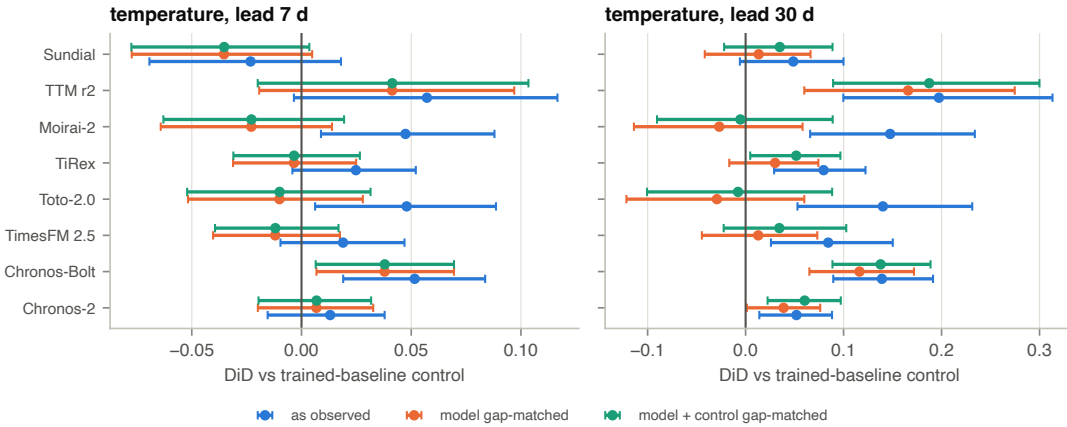


Figure 6: Difference-in-differences against the trained-baseline control, with 95% intervals. Positive values mean the model loses more skill in the tropics than the control does. Masking the control as well as the model is the like-for-like comparison.

5.5 Skill reverses by monsoon regime

Aggregate leaderboards hide the benchmark’s sharpest result. Stratifying July–August targets by spell, and blocking intervals by spell *event* rather than station-day, 61 of 72 contrasts are significant. In break spells every model is better than climatology, and the foundation models hold the widest margins: Chronos-Bolt reaches +0.440 on temperature at a 30-day lead, against PatchTST’s +0.106. In active spells the ordering inverts. Every model falls below climatology at 7- and 30-day leads, and the foundation models lose the most ground: Chronos-Bolt drops to −0.989, against PatchTST’s −0.193. Humidity repeats the pattern. Forecast quality is therefore lowest during the very episodes that drive flooding, transplanting and evacuation decisions.

5.6 Calibration and compute

Calibration varies more than accuracy. TimesFM 2.5 (0.831), Toto-2.0 (0.826) and TiRex (0.797) are close to the nominal 80 %; Chronos-2, the most accurate model overall, is sharp on rainfall (0.629); and Sundial is severely under-dispersed (0.323). A GPU diagnostic confirms the last is a property of the model rather than of our adapter. Interval width is flat in the number of sampled trajectories: 1.44, 1.56 and 1.59 for 20, 100 and 500 samples. The output shape matches the documented layout. And the trajectory spread of 0.64°C sits against the model’s own RMSE of 1.51°C .

The whole zero-shot leaderboard is cheap to reproduce. Every model fits in 4.2 GiB of GPU memory. All 8 models over all 16 tasks cost 4.16 GPU-hours on a Tesla T4, ranging from 0.6 minutes for TTM r2 to 130 minutes for Sundial. By contrast AutoARIMA alone costs about 20.7 CPU-hours. Accuracy does not track size: TiRex ranks second overall with 35.3M parameters and 568 MiB, while Toto-2.0 (312.7M) ranks fourth and Sundial (128.3M) eleventh (Figure 7).

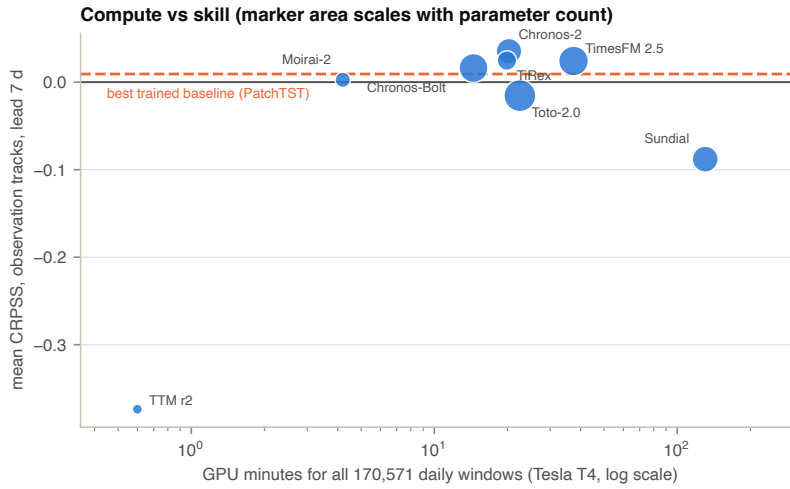


Figure 7: Compute against skill on the observation tracks at a 7-day lead; marker area scales with parameter count. The full leaderboard is reproducible in roughly four GPU-hours on a single GPU.

6 Discussion

What the nulls mean. A reader could summarise this paper as “foundation models mostly tie”. That summary is accurate and is the point. The zero-shot claim is usually evidenced on aggregate leaderboards without significance testing. Score the same models on a tropical station network, resample at the forecast origin, and correct across 48 comparisons, and most margins on observations become indistinguishable from a well-built baseline. The models are not weak: six of eight outrank every trained baseline, and 272 of 432 skill intervals exclude climatology. But the size of the advantage on real station data is smaller than the literature’s framing implies.

Why rainfall is different. The rainfall result is not a modelling failure so much as a property of the target. Daily tropical rainfall is zero-inflated (67.6 % dry station-days) and convective, so beyond a day the predictable signal is largely the seasonal cycle, which a day-of-year climatology already encodes. A forecast that reproduces the climatological distribution is close to the achievable ceiling, and no model in our roster exceeded it beyond a one-day lead. This is worth

stating plainly because rainfall is the variable on which the operational value of these models would be judged in Bangladesh.

Why regime stratification matters. The spell result is the clearest argument against aggregate reporting. The same models are strongly skilful during break spells and significantly worse than climatology during active spells, a reversal that vanishes in a seasonal average. One mechanism would explain this, though we do not test it. A model with a long context window carries the persistent warm, dry conditions of a break forward. That is rewarded while the break holds, and penalised as soon as convection resumes. Whatever the cause, a benchmark that reports only aggregate skill would certify these models as adequate for exactly the situation in which they are least reliable.

Implications for benchmark design. Three of our results are really statements about how evaluation is done. Scoring against reanalysis inflates apparent skill and, more subtly, inflates the *margin* between model classes, so reanalysis-only benchmarks are not a neutral convenience. Stratifying by a physically defined regime exposes failures that aggregates hide. And a context-availability control changes the interpretation of a geographic gap: without masking the trained baseline identically, our own earlier reading, that longer context dissolves the tropical deficit, did not survive.

Affordability. The entire zero-shot leaderboard reproduces in 4.16 GPU-hours on a single GPU, and the strongest classical baseline we ran costs about five times that in CPU time. For the research communities this benchmark is about, zero-shot models are already the affordable option; the question this paper answers is what that affordability currently buys on their own data.

7 Limitations

The equity contrast rests on ten stations. The tropical-versus-temperate comparison uses the 10 GHCN-Daily stations in Bangladesh, giving 1,154 rainfall and 1,478 temperature windows, an order of magnitude fewer than the BMD track. Every equity figure in this paper should be read with that n attached. Three design choices limit but do not eliminate the risk: the temperate reference is drawn from the same instrument network; the gap-matched control is applied to models and baselines alike; and intervals come from a moving-block bootstrap over origins rather than from window counts. We report the contrast because it is the only like-for-like comparison available, not because ten stations settle the question.

One country, one climate. Findings are specific to Bangladesh’s monsoon regime. The protocol, controls and significance machinery transfer; the numbers do not. Whether the long-lead temperature deficit appears in other tropical networks is open.

Zero-shot only. We evaluate models as they are shipped. Few-shot adaptation and light fine-tuning are out of scope; they are the obvious next step, and cheap for several models in the roster. This paper therefore bounds the zero-shot claim rather than the models’ ceiling.

The roster will age. Eight models current in 2026 will not be current for long. The harness, cached forecasts and scoring code are released so that adding a model is a single run against unchanged windows.

Imputation in the source record. The published BMD data was imputed by its authors before release. We detect and flag suspect values and exclude flagged targets from scoring, but detection is imperfect; flagged shares over the test period run from 0.08 % (rainfall) to 4.44 % (humidity), with one station-variable at 14.1 %.

Two comparisons are weaker than the rest. Bright sunshine hours and downward short-wave radiation are not the same physical quantity, so that row of the observation-versus-reanalysis contrast is reported but not relied on. The CHIRPS dekadal task is an independent-source check at coarse resolution, not a headline.

Mechanisms are hypothesised, not tested. We measure the regime reversal and the context-availability effect; we do not identify the architectural cause of either.

8 Conclusion

We built a benchmark for daily weather forecasting on tropical-monsoon station observations, with the controls needed to interpret a tropical result: a same-network temperate reference, a reanalysis track at identical coordinates, and a gap-matched context control applied to models and baselines alike. We evaluated 8 zero-shot time-series foundation models against 10 trained and statistical baselines over 170,571 rolling-origin windows, treating the forecast origin as the unit of resampling.

The picture is mixed. Foundation models are the strongest family, but on station observations most of their advantage is not statistically separable. Nothing beats a day-of-year climatology on tropical rainfall beyond a one-day lead; reanalysis flatters every model relative to the stations it represents; and skill reverses between monsoon break and active spells. The long-lead tropical temperature deficit that half the roster shows is mostly, though not entirely, a consequence of sparser station context rather than of the tropics.

For practitioners in data-sparse regions the practical reading is twofold. Zero-shot models are already affordable and already competitive on smooth variables; for rainfall, a well-built climatology remains the thing to beat. For benchmark builders the reading is that scoring target, regime stratification and context availability are not implementation details: each one changed a conclusion in this study.

Data and code availability

The benchmark definition and window index, the evaluation harness, the significance code, the figure scripts and the aggregated score tables behind every number in this paper are available at <https://github.com/ihpolash/BanglaWeatherBench>. The exact snapshot this paper describes is archived as v1.0.1 at <https://doi.org/10.5281/zenodo.22825319>. Source data are redistributed only where licensing permits: the BMD station record is CC-BY-4.0 [35]; GHCN-Daily and NASA POWER are public domain; CHIRPS is open [10]. Derived tracks are rebuilt by the released pipeline rather than shipped, as are the per-window forecast cache and per-window scores, which run to several gigabytes. The release includes a datasheet in the sense of Gebru et al. [11], covering composition, collection, preprocessing, uses and limitations.

Reproducibility

Model inference runs in per-model isolated environments; every forecast file is verified for row counts, window identity, lead coverage, missing values, quantile monotonicity and non-negativity before it enters a leaderboard. The headline numbers in this paper are recomputed from released

artefacts by `scripts/audit_claims.py`, which runs against the committed score tables and window index without rebuilding the pipeline.

Declaration on the use of generative AI

The author used an AI assistant (Claude, Anthropic) for implementation support, analysis and figure scripting, and for drafting and editing this manuscript. All of that work was carried out under the author’s direction and review. The research questions, benchmark design, evaluation protocol and the interpretation of the results are the author’s. All numerical results were produced by the released code and were recomputed from the released artefacts by an audit script before submission; every reference was verified by the author against its primary source. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

References

- [1] Colin Aitken, Rajat Masiwal, Adam Marchakitus, Katherine Kowal, Mayank Gupta, Tyler Yang, Amir Jina, Pedram Hassanzadeh, William R. Boos, and Michael Kremer. Designing probabilistic AI monsoon forecasts to inform agricultural decision-making. *arXiv preprint arXiv:2603.07893*, 2026.
- [2] Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.
- [3] Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. arXiv:2403.07815.
- [4] Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, et al. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2026.
- [5] Andreas Auer, Patrick Podest, Daniel Klotz, Sebastian Böck, Günter Klambauer, and Sepp Hochreiter. TiRex: Zero-shot forecasting across long and short horizons with enhanced in-context learning. *arXiv preprint arXiv:2505.23719*, 2025.
- [6] Ben Cohen, Emaad Khwaja, Youssef Doubli, Salahidine Lemaachi, Chris Lettieri, Charles Masson, Hugo Miccinilli, Elise Ramé, Qiqi Ren, Afshin Rostamizadeh, and Jean Ogier du Terrail. This time is different: An observability perspective on time series foundation models. *arXiv preprint arXiv:2505.14766*, 2025.
- [7] Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. *arXiv preprint arXiv:2310.10688*, 2024.
- [8] Vijay Ekambaram, Arindam Jati, Pankaj Dayama, Sumanta Mukherjee, Nam H. Nguyen, Wesley M. Gifford, Chandra Reddy, and Jayant Kalagnanam. Tiny time mixers (TTMs): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series. *arXiv preprint arXiv:2401.03955*, 2024.
- [9] Zannatul Ferdoushi, D. A. Quadir, and S. M. Quamrul Hassan. Active and break spells of summer monsoon over Bangladesh. *Heliyon*, 2023. doi: 10.1016/j.heliyon.2023.e20347.

- [10] Chris Funk, Pete Peterson, Martin Landsfeld, Diego Pedreros, James Verdin, Shradhanand Shukla, Gregory Husak, James Rowland, Laura Harrison, Andrew Hoell, and Joel Michaelsen. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Scientific Data*, 2:150066, 2015. doi: 10.1038/sdata.2015.66.
- [11] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.
- [12] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [13] Rakshitha Godahewa, Christoph Bergmeir, Geoffrey I. Webb, Rob J. Hyndman, and Pablo Montero-Manso. Monash time series forecasting archive. *arXiv preprint arXiv:2105.06643*, 2021. NeurIPS Datasets and Benchmarks.
- [14] Aman Gupta, Aditi Sheshadri, and Dhruv Suri. MAUSAM: An observations-focused assessment of global AI weather prediction models during the south asian monsoon. *Journal of Advances in Modeling Earth Systems*, 2026. doi: 10.1029/2025MS005568. arXiv:2509.01879.
- [15] Tao Han, Zhibin Wen, Zhenghao Chen, Dazhao Du, Song Guo, and Lei Bai. Benchmarking physics-informed time-series models for operational global station weather forecasting. *arXiv preprint arXiv:2406.14399*, 2026. ICML 2026.
- [16] Weixin Jin, Jonathan Weyn, Pengcheng Zhao, Siqi Xiang, Jiang Bian, Zuliang Fang, Haiyu Dong, Hongyu Sun, Kit Thambiratnam, and Qi Zhang. WeatherReal: A benchmark based on in-situ observations for evaluating weather models. *arXiv preprint arXiv:2409.09371*, 2024.
- [17] Usman Gani Joy, Shahadat Kabir, and Tasnim Niger. Attention-enhanced LSTM modeling for improved temperature and rainfall forecasting in bangladesh. *arXiv preprint arXiv:2510.10702*, 2026.
- [18] Zhe Li, Xiangfei Qiu, Peng Chen, et al. TSFM-Bench: A comprehensive and unified benchmark of foundation models for time series forecasting. *arXiv preprint arXiv:2410.11802*, 2025. KDD 2025.
- [19] Chenghao Liu, Taha Aksu, Juncheng Liu, Xu Liu, Hanshu Yan, Quang Pham, Silvio Savarese, Doyen Sahoo, Caiming Xiong, and Junnan Li. Moirai 2.0: When less is more for time series forecasting. *arXiv preprint arXiv:2511.11698*, 2026. Salesforce AI Research.
- [20] Yong Liu, Guo Qin, Zhiyuan Shi, Zhi Chen, Caiyin Yang, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Sundial: A family of highly capable time series foundation models. *arXiv preprint arXiv:2502.00816*, 2025.
- [21] Somnath Luitel, Parthasarathi Mukhopadhyay, et al. GraphCast skill and systematic biases in indian summer monsoon forecasts: Evaluation against ERA5 and IMERG. *arXiv preprint arXiv:2607.11905*, 2026.
- [22] Nick Masi and Randall Balestriero. SAFE: A novel approach to AI weather evaluation through stratified assessments of forecasts over Earth. *arXiv preprint arXiv:2510.26099*, 2025.

- [23] Amy McGovern, Taylor Mandelbaum, Daniel Rothenberg, Nicholas Loveday, Corey Potvin, Montgomery Flora, Linus Magnusson, Eric Gilleland, and John Allen. Extreme weather bench: A framework and benchmark for evaluation of high-impact weather. *arXiv preprint arXiv:2605.01126*, 2026.
- [24] Matthew J. Menne, Imke Durre, Russell S. Vose, Byron E. Gleason, and Tamara G. Houston. An overview of the global historical climatology network-daily database. *Journal of Atmospheric and Oceanic Technology*, 29(7):897–910, 2012. doi: 10.1175/JTECH-D-11-00103.1.
- [25] Marcel Meyer, Sascha Kaltenpoth, Kevin Zalipski, and Oliver Müller. Rethinking evaluation in the era of time series foundation models: (Un)known information leakage challenges. *arXiv preprint arXiv:2510.13654*, 2026.
- [26] Amirpasha Mozaffari, Amanda Duarte, Lina Teckentrup, Stefano Materia, Gina E. C. Charnley, Lluís Palma, Eulalia Baulenas Serra, Dragana Bojovic, Paula Checchia, Aude Carreric, and Francisco Doblas-Reyes. The rise of AI in weather and climate information and its impact on global inequality. *arXiv preprint arXiv:2603.05710*, 2026.
- [27] Juan Nathaniel, Yongquan Qu, Tung Nguyen, Sungduk Yu, Julius Busecke, Aditya Grover, and Pierre Gentine. ChaosBench: A multi-channel, physics-based benchmark for subseasonal-to-seasonal climate prediction. *arXiv preprint arXiv:2402.00712*, 2024.
- [28] Zhongzheng Qiao, Sheng Pan, Anni Wang, Viktoriya Zhukova, Yong Liu, Xudong Jiang, Qingsong Wen, Mingsheng Long, Ming Jin, and Chenghao Liu. It’s TIME: Towards the next generation of time series forecasting benchmarks. *arXiv preprint arXiv:2602.12147*, 2026. ICML 2026.
- [29] Xiangfei Qiu, Jilin Hu, Lekui Zhou, Xingjian Wu, et al. TFB: Towards comprehensive and fair benchmarking of time series forecasting methods. *arXiv preprint arXiv:2403.20150*, 2024. PVLDB.
- [30] Stephan Rasp, Stephan Hoyer, Alexander Meroze, Ian Langmore, Peter Battaglia, Tyler Russell, Alvaro Sanchez-Gonzalez, Vivian Yang, Rob Carver, Shreya Agrawal, Matthew Chantry, Zied Ben Bouallegue, Peter Dueben, Carla Bromberg, Jared Sisk, Luke Barrington, Aaron Bell, and Fei Sha. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *arXiv preprint arXiv:2308.15560*, 2024.
- [31] Wolfgang R. Rowell Jr. and Lucas S. Kupssinskü. Performance evaluation of GraphCast for medium-range weather forecasting over brazil. *arXiv preprint arXiv:2606.06348*, 2026.
- [32] Yingshuo Wang, Xian Sun, Lingdong Kong, Wei Gao, Yanhang Li, Zhichao Fan, and Zexin Zhuang. Do time series foundation model benchmarks hide regime-dependent failures? evidence from traffic speed forecasting. *arXiv preprint arXiv:2606.18367*, 2026.
- [33] Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning*, 2024. arXiv:2402.02592.
- [34] Yifang Zhang, Shengwu Xiong, Henan Wang, Wenjie Yin, Jiawang Peng, Yuqiang Zhang, Chen Zhou, Hua Chen, Qile Zhao, and Pengfei Duan. How effective are time-series models for precipitation nowcasting? a comprehensive benchmark for GNSS-based precipitation nowcasting. *IEEE Transactions on Geoscience and Remote Sensing*, 2025. arXiv:2509.25263.
- [35] Md Zubair, Md. Nafiz Ishtiaque Mahee, Khondaker Masfiq Reza, MD Shahidul Salim, and Nasim Ahmed. High volume real-world weather data, 2024. CC BY 4.0.